

Saad Abdur Razzaq

Machine Learning Researcher | Optimization & Training Dynamics | Computer Vision

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[LinkedIn](#) | [GitHub](#) | [YouTube](#) | [The Applied Engineer SubStack](#)

Personal Summary

Machine learning researcher interested in optimization, training dynamics, and the mechanisms underlying learning and generalization in deep neural networks. My work involves implementing learning algorithms and neural architectures from scratch, systematically comparing model architectures, and analyzing computational and predictive efficiency, with a particular focus on computer vision and visual representation learning. I design controlled experiments to investigate model behavior, test hypotheses, quantify performance differences, and communicate findings through reproducible analysis and visualization.

Research Interests

Core: Deep Learning Optimization, Training Dynamics, Architecture Analysis, Model Efficiency, Computer Vision

Methods: From-Scratch Implementation, Systematic Comparison, Quantified Evaluation, Visualization

Education

National University of Computer and Emerging Sciences (FAST–NUCES)

B.Sc. Computer Science

Sep 2020 – Jun 2024

Chiniot–Faisalabad, Pakistan

- Relevant coursework: Artificial Intelligence, Fundamentals of Computer Vision, Applied Machine Learning, Numerical Computing, Probability and Statistics

Research Experience

Adaptive RMSprop+: Enhanced Optimization Framework for Deep Learning

[GitHub](#)

- Designed and implemented a novel extension of the RMSprop optimizer incorporating dynamic epsilon adjustment, cyclical learning rates with warmup, adaptive gradient noise, and layer-wise momentum adaptation.
- Achieved 28% faster convergence, 60% improvement in training stability, and 1.5% absolute accuracy gain over standard RMSprop across image classification benchmarks.
- Implemented the optimizer from scratch in PyTorch, including mathematical formulation of gradient-norm-aware epsilon scaling and cosine-annealed cyclical learning rate schedules.
- Conducted systematic comparison against Adam and standard RMSprop, demonstrating superior convergence across CIFAR-10 and language modeling tasks.

From-Scratch Recurrent Neural Network for Sequential Arithmetic Computation

[GitHub](#)

- Implemented a vanilla RNN entirely from first principles using NumPy, including manual forward propagation, backpropagation through time, gradient clipping, and numerical gradient checking.
- Investigated the ability of recurrent architectures to learn symbolic arithmetic operations (cumulative products) through temporal dependencies, demonstrating successful convergence on sequential computation tasks.
- Validated implementation correctness through numerical gradient checking, ensuring analytical gradients matched numerical approximations to high precision.
- Analyzed training dynamics, gradient propagation, and weight initialization to understand how RNNs represent and learn sequential computation.

Memory-Augmented Reinforcement Learning: A Systematic Comparison Framework

[GitHub](#)

- Built a modular experimental framework for systematically comparing RL agents with and without explicit memory mechanisms across multiple decision-making tasks.
- Implemented memory-augmented Q-learning and DQN agents from scratch to isolate the impact of memory on learning dynamics.
- Demonstrated 63% performance improvement with memory augmentation in T-Maze task and 56% in spatial alternation, with evidence of cross-task transfer benefits.
- Developed visualization and analytics tools for learning curves, memory utilization patterns, and transfer learning analysis.

Quantization and Efficiency Analysis of Transformer-Based Sequence Models

[GitHub](#)

- Built an experimental framework for systematically evaluating the efficiency-quality trade-off in transformer-based neural machine translation under INT8 quantization.
- Quantized Helsinki-NLP OPUS-MT model using ONNX Runtime, achieving 4× model size reduction, 2.7× inference latency improvement, and 4× memory reduction.
- Maintained translation quality within 1.2% BLEU score of the original FP32 model, analyzing layer sensitivity and calibration strategies for preserving quality under aggressive compression.

Robustness Evaluation of FaceNet-Based Recognition Systems

[GitHub](#)

- Developed a modular facial recognition pipeline using PyTorch, MTCNN detection, and FaceNet/InceptionResNetV1 embeddings.
- Evaluated model robustness under systematic distribution shifts including brightness variation, expression changes, image quality degradation, and age variance.
- Achieved 95.2% accuracy on standard conditions and systematically analyzed failure modes through confusion matrices, ROC curves, and confidence calibration.

Selected Projects

Synthetic Sequence Generation with LSTM Networks

[GitHub](#)

- Developed an LSTM-based sequence-to-sequence framework for generating realistic synthetic sequences from structured sequential data.
- Learned concept embeddings using skip-gram architecture with negative sampling to capture semantic relationships between features.
- Applied the framework to generate synthetic sequences that preserve statistical properties while preventing re-identification.

Deep Q-Network for CartPole

[GitHub](#)

- Implemented DQN from scratch in PyTorch with experience replay and epsilon-greedy exploration to solve CartPole-v1.
- Achieved convergence within 200–500 episodes, demonstrating stable learning through replay buffer and target network updates.

Final Year Project

GynaeAutoCare – Digital Healthcare Platform for Women’s Health

FAST–NUCES

- Designed and developed a full-stack healthcare system using Django, PostgreSQL, Celery, and Django Channels.
- Implemented electronic health records, telemedicine infrastructure, appointment scheduling, and clinical decision support.

Industry Experience

Software Developer (Full Stack)

July 2024 – Present

ProActive Enterprises (Private) Limited

Remote, Pakistan

- Developing full-stack applications across frontend, backend, and database layers; gained experience with production systems, authentication, and integration workflows.

Teaching Assistant – Data Science & Business Analytics

Aug 2023 – Jun 2024

FAST–NUCES, Chiniot–Faisalabad Campus

Pakistan

- Supported student learning in Python-based data analysis, machine learning, & predictive modeling through tutorials, technical guidance, & code reviews; helped students debug implementations & develop high-quality solutions.

Technical Skills

ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, ONNX Runtime, FastAI

Architectures: CNNs, RNNs, LSTMs, DQN, VGG, FaceNet, Vision Transformers (ViT), Seq2Seq, ResNET

Techniques: Optimization Algorithms, Model Quantization, Transfer Learning, Knowledge Distillation, Regularization

Languages: Python, C, C++, R, SQL

Tools: NumPy, Pandas, OpenCV, Git, Docker, PostgreSQL, FastAPI

Selected Certifications

- IBM – AI Developer Professional Certificate, 2025
- IBM – Generative AI Engineering Professional Certificate, 2025
- Google – Advanced Data Analytics Professional Certificate, 2025

Knowledge Sharing

Founder & Author – The Applied Engineer

[Substack](#)

- Publishing daily articles that demystify the mathematical foundations of machine learning like loss, gradients, backpropagation, and the training loop — for readers with zero prior knowledge.
- Filling a gap in ML education by grounding every concept in practical, real-world examples and step-by-step reasoning, rather than abstract formulas or unexplained code.
- Demonstrating deep understanding of model internals and training dynamics by translating mathematically rigorous concepts into accessible, intuitive explanations.

Languages

English: IELTS Academic 7.0 overall (no band below 6.5) | **Urdu:** Native